

THE KING'S GAUNTLET

Live standings platform — opening-day traffic, edge & database load, incidents, and forward capacity. Hosted by Hera.

Event day 2026-06-01

Window 15:00–04:00 UTC

11k visitors

148k page views

8.42M API requests

Status **recovered**, 0 data loss

EXECUTIVE SUMMARY

Opening day drew ~11k unique visitors and 148k page views, peaking at ~2,590 unique viewers in the 17:00 UTC hour. The platform absorbed it but hit its engineered ceilings during the peak, surfacing three same-day, viewer-facing incidents — all configuration/edge issues, all fixed by 20:00 UTC with no data loss. The edge and database-throughput layers have enormous headroom; the binding constraint for growth is long-lived SSE connections at the API tier.

11K

Unique visitors

13.4k incl. bots
(Cloudflare)

148K

Page views

~13.2k sessions

~400

Peak concurrent
viewers

17:11 UTC marquee
spike

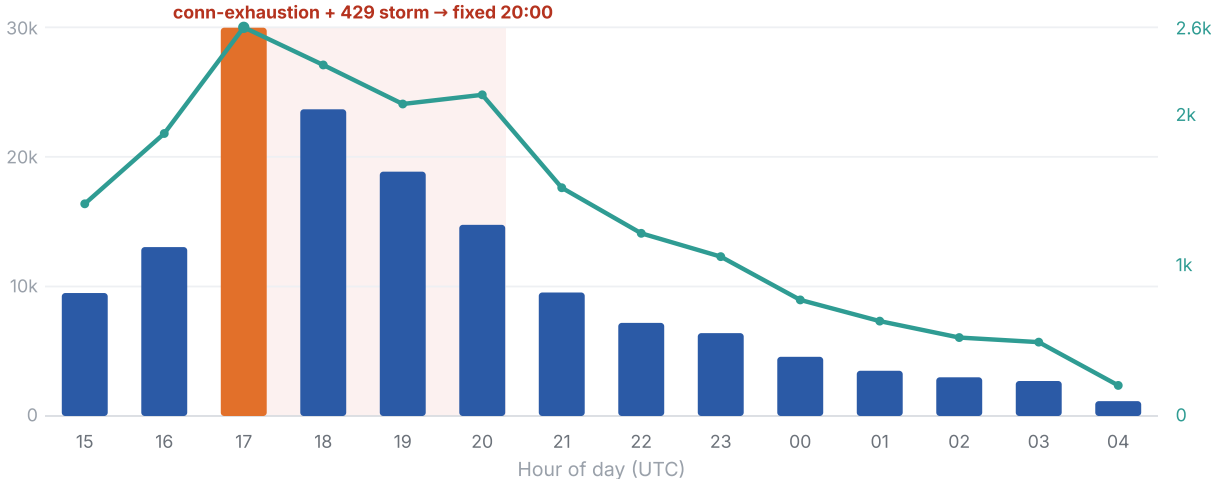
86%

API edge cache-hit

coalescing working

AUDIENCE & TRAFFIC

Near-zero until the 15:00 UTC pre-show, then a sharp ramp to a 17:00 peak and a long, well-behaved decay through to ~04:00. Page views fall faster than unique viewers through the evening — the audience tunes in for marquee moments and drifts.



■ Page views / hr (left axis) ■ Peak hour — 29,967 ■ Unique viewers / hr (right axis)

Source: PostHog \$pageview, project hera-streamer-invitational-2026. The 7-minute edge outage (17:31-17:38) sits inside the 17:00 bucket.

Top countries — by API requests, 24h

United States		2.06M
Germany		1.09M
Canada		535k
Sweden		368k
United Kingdom		365k
Netherlands		317k
Poland		230k

A genuinely global audience — Cloudflare zone analytics.

Session shape

13.2K

Sessions

2,590

Peak unique / hr

410

Peak unique / min
17:11 UTC

0

Uncaught FE
errors
\$exception, all day

The front end degraded gracefully — retries, not crashes.

EDGE & CACHING — Cloudflare

The edge did the heavy lifting. Of 8.42M requests to the API hostname, ~86% were served from Cloudflare's cache — origin saw only the remaining ~14% (~15–23 req/s). The front-end Worker that caused the kickoff outage is now on a paid plan with vast headroom.

API edge cache breakdown — 8.42M requests, 24h



■ Hit — 7.27M ■ Expired (revalidated 304) — 429k ■ None / uncacheable — 380k

■ Bypass (admin cookie) — 248k ■ Dynamic — 52k

The **Bypass** slice confirms the admin cookie-bypass cache rule is live; **Hit + Expired (304)** = ~91% of traffic never required a full origin response.

Front-end edge Worker — criticalbit-router, 24h

714K

Worker requests
vs 10M/mo
included

0.75MS

Median CPU / req

9.01M

Zone requests
86.9% cached

29 GB

Served (90%
cached)

The free-tier 100k/day cap that 1027'd the site at kickoff is gone; 5× launch ≈ \$5–30/mo.

Edge status codes — API, 24h

200 OK		5.10M
304 Not Modified		2.73M
499 Client Closed		380k
429 Rate-limited		178k
500 Server Error		34k

The 429 (limiter storm) and 500 (connection exhaustion) counts are the incident window — both → 0 after 20:00.

ORIGIN & DATABASE LOAD — GCP Cloud Monitoring

The supply side tells the real capacity story. At the peak the API ran **at its engineered ceiling** — 20 of 20 Cloud Run instances and ~800 of 800 concurrent requests per instance — and the database saturated its (then) 100-connection cap. Notably, database CPU never broke 46%: the binding resource was **connection & request slots**, not throughput.

Peak utilization vs engineered ceiling

Cloud Run instances		20 / 20
Per-instance conc...		~800 / 800
DB connections (p...		97 / 100*
DB conns on deploy		166 / 200
DB CPU		46%

*Cap was raised 100 → 200 at 20:00 UTC (#173). Red = saturated, amber = high, green = comfortable. CPU staying low is the proof that read-throughput was never the constraint.

Why a viewer-count model understated the load by ~20×

In the clean post-incident wind-down, the API still held ~10 of 20 instances at 02:00 UTC when only ~276 viewers were *active*. The reason: every open tab holds a Server-Sent-Events connection (`/v1/stream`, ~249k connections in 24h) that occupies a server request slot for up to an hour — watching or not. Capacity is bound by **cumulative open tabs**, not instantaneous viewers.

INCIDENT TIMELINE 2026-06-01 UTC

Three viewer-facing incidents, all configuration/edge — not raw capacity — and all resolved same-day with no data loss.

- 15:00** — Pre-show ramp begins; traffic climbs from a standing start.
- 17:10–18:40** — Peak. API pinned at 20/20 instances; Cloud SQL connections climb to the 100 cap (state-revision accumulation) → intermittent 5xx.
- 17:31–17:38** — **Edge outage (7 min)**. The front-end Worker hits Cloudflare's free 100k-req/day cap (error 1027) — full site-shell blackout. Visible as a 7-minute hole in page views.
- 18:50–19:50** — **429 storm**. Rate limiter keyed on Cloudflare's edge IP + Vary: Cookie defeating CDN coalescing → ~80% of reads rejected; 4xx peaks at ~99k / 10 min. Client retries (`standings.retry`) hit 951/hr.
- 20:00** — **Resolved**. DB tier bump to 200 connections (#173) + limiter/cache fix (#176) deploy. 429s, 5xx, and retries collapse to ~0.
- 20:10–21:25** — Zero-downtime schema re-key (#179/#181) and stats windowing (#185) land cleanly under live traffic — 0× 5xx through both rollovers.

Fingerprints independently confirmed across PostHog (page-view blackout + retry spike), Cloudflare (429/500 counts), and Cloud Monitoring (instance + connection saturation).

CAPACITY & FORWARD HEADROOM

Huge headroom — not the constraint

Edge / Worker: 714k/day on a 10M/mo plan.

DB read throughput: ~15–23 req/s origin, CPU ≤46%, 86% edge-cached.

The pre-event model's headline fear — "DB read saturation" — never materialized.

The binding constraint — ~1.5–2× organic

SSE request slots (Cloud Run concurrency × instances) and **DB connection slots** (pool × instances × revisions). Both are long-lived-connection limits — they were *at* the ceiling at launch. Finals (potentially 2–5× launch) need the changes below.

ACTION ITEMS

Eight issues filed across the three repositories, prioritized by leverage. The SSE-stickiness fix attacks the root cost and is the cheapest, highest-impact change.

PRI	ACTION	REPO	ISSUE
P1	Drop the SSE connection on hidden tabs. Attacks the dominant capacity cost (abandoned tabs holding seats).	web	#285
P1	Instrument the live SSE subscriber count. The one missing ground-truth number for capacity planning.	api	#194
P2	PgBouncer connection pooling. Decouples app connections from the DB cap; prerequisite for scaling instances.	api	#196
P2	Raise Cloud Run maxScale 20 → 40. Doubles the seat ceiling (gated on pooling).	api	#195
P3	Raise per-instance concurrency 800 → 1200 after a memory stress-test. Free on the DB side.	api	#197
P3	Deploy-freeze during marquee matches — DB connection spike on rollover.	api	#198
P3	Deploy-freeze — CDN cache purge + stale code-split chunks.	web	#286
P3	No Worker deploys at peak; wire Workers Builds (close merge≠deploy).	router	#15

The King's Gauntlet — Launch Day Metrics. Prepared 2026-06-02. Figures are 24h aggregates for the 2026-06-01 event window unless noted; SSE/active-viewer figures are instantaneous peaks.

Demand: PostHog (hera-streamer-invitational-2026) **Edge:** Cloudflare (criticalbit.gg)

Supply: GCP Cloud Monitoring (Cloud Run + Cloud SQL)